

Yale Xie

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EDUCATION

Yale University

B.S. Computer Science and Mathematics

2026 – 2030

New Haven, CT

Liberal Arts and Science Academy

High School Diploma

2022 – 2026

Austin, TX

- **GPA:** 4.66/4.00; **SAT:** 1570/1600
- **Awards:** FBLA National Champion (2025), WPI Innovation Award (2025)

PROJECTS

Yale Events

September 2026 – Present

- Combined 21 Yale event calendars into one searchable collection of over 2,000 events by building a Python web-scraping pipeline that feeds into a SQLite database
- Removed 230+ duplicate listings by writing a fuzzy-matching dedupe function that considers title similarity, start times, and location agreement
- Shipped a public FastAPI service with Google Calendar integration (one-click add and automatic subscriptions), earning a \$200 bounty from the Yale College Council Tech Division

EDGAR Intelligence

June 2026 – August 2026

- Built and deployed a full-stack RAG application indexing over 8,000 SEC filing chunks across 13 public companies using FastAPI, Pinecone, OpenAI embeddings, Cohere reranking, a Render backend, and a Vercel frontend
- Improved retrieval quality using top-50 Pinecone ANN search, Cohere cross-encoder reranking, query expansion, and lexical boosting, achieving 91% RAGAS answer faithfulness on factual queries
- Implemented a CI/CD pipeline with GitHub Actions, running 152 network-free unit tests on code changes and a 110-case regression suite that blocks merges below a 97% pass rate threshold

EXPERIENCE

Quantitative Research Intern

June 2025 – August 2025

The Cake Shop Capital

Boston, MA

- Co-developed a GPT-4o-based NLP pipeline to extract and quantify legal risk changes across sequential SEC 10-K and 10-Q filings, creating an investment signal with a 51.4% backtested cumulative return between 2020 and 2025
- Backtested the trading signal in Python across multiple risk metrics (Fama-French regression, Sharpe ratio, volatility, win/loss ratio), confirming 5.8% annualized statistically significant alpha at 6% annualized volatility

Research Intern

February 2025 – July 2025

Davidson College

Remote

- Built and optimized a statistical model in Python with Bayesian Additive Regression Trees (BART) to assess the relative importance of the Fama-French factors in U.S. large-cap equity returns
- Achieved an R^2 of 0.87 versus 0.22 for the OLS baseline, demonstrating the model's ability to capture nonlinear relationships and interactions among factors
- Authored a peer-reviewed [research article](#) published in the *Pioneer Research Journal*

Data Analytics Intern

February 2025 – July 2025

Kelleher Consulting Group

Remote

- Analyzed IPEDS data using Pandas, NumPy, and Matplotlib to size and segment target markets for WorldQuant University's go-to-market strategy
- Benchmarked competing data-science education products using capability matrices and presented positioning recommendations to the executive team

SKILLS AND INTERESTS

Languages: Python, Java, Racket

Frameworks/Libraries: Pandas, NumPy, PyTorch, scikit-learn, SciPy, PyMC, Matplotlib, FastAPI

Tools/Platforms: Git, GitHub Actions (CI/CD), CLI, Pinecone, Chroma, Cohere, Vercel, Render, REST APIs

Methods: Bayesian modeling, regression, neural networks, retrieval-augmented generation (RAG)

Interests: Poker, chess, tennis, pickleball, ping pong, Roman history, cooking, classical music, The Weeknd